Abstract Algebra - Homework 2

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Problem 1

(1)
$$ij = k, \quad ij = -ji \implies -ji = k \implies ji = -k$$

Hence, $ij \neq ji$, so multiplication is not commutative and $\mathbb{H}$ is a non-commutative ring.

(2) **(a)** Let $x, y \in \mathbb{H}$. One can verify by direct computation that conjugation reverses the order of multiplication, $\overline{xy} = \bar{y}\bar{x}$. Note that for any $h = a + bi + cj + dk \in \mathbb{H}$, we have

$$N(h) = h\bar{h} = a^2 + b^2 + c^2 + d^2 \in \mathbb{R},$$

so $N(h)$ lies in the center of $\mathbb{H}$ and commutes with all elements. Using this, we compute:

$$\begin{aligned} N(xy) &= (xy)\overline{(xy)} \\ &= (xy)(\bar{y}\bar{x}) \\ &= x(y\bar{y})\bar{x} \\ &= xN(y)\bar{x} \\ &= N(y)x\bar{x} \\ &= N(y)N(x) = N(x)N(y). \end{aligned}$$

(b) Since $N(x) = x\bar{x} = a^2 + b^2 + c^2 + d^2$, it follows that

$$N(x) = 0 \iff a = b = c = d = 0 \iff x = 0.$$

Thus, $N(x) = 0$ if and only if $x = 0$.